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Nitte (Deemed to be University)

(Established under Sec (3) of UGC ACT,1956)

First Semester MCA (CBCS) Degree Examinations, Dec. 2025 - Jan. 2026

25MCA101 - DATA STRUCTURES WITH ALGORITHMS

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Define an Array. Explain how a two-dimensional array is represented in memory, and describe the method used to calculate the address of an element in a two-dimensional array.	10	L3	1	2
	b)	Define Abstract Data Type.*Write an ADT for rational numbers.	10	L2	1	1
2.	a)	With a neat diagram explain algorithm development process.	10	L2	1	1
	b)	Distinguish between (i) Algorithm and pseudo code. (ii) Big oh and omega notation	10	L2	1	1
		Unit – II				
3.	a)	Define Singly linked list. Write C implementation code for the following: i) Delete all the occurrences of a given number from the singly linked list. ii) Insert a node into a singly linked list after a given node.	10	L4	2	2
	b)	Compare and contrast Doubly linked list and Singly linked list. Write the C implement code for realization of Singly linked list	10	L3	2	2
4.	a)	Define Ordered Linked list. Write C implementation code for the following: i) Delete a node (whose info field is specified) from an ordered linked list. ii) Insert a node into an ordered linked list.	10	L4	2	2
	b)	Write the C implement code for realization of Singly linked linked list	10	L3	2	2
		Unit – III				
5.	a)	Define Stack. Explain the primitive operations that can be performed on stack and give the C implementation code for the operations.	10	L2	3	1
	b)	Write an algorithm to evaluate an expression given in postfix notation. Trace your algorithm for the following expression AB^CD-EF/GH+/*. A=2, B=2, C=6, D=2, E=12, F=2, G=2, H=1.	10	L3	3	2
6.	a)	List the difference between ordinary queue and circular queue. Write the implementation details of primitive functions of circular queue(Static).	10	L3	3	2
	b)	Describe the Towers of Hanoi Problem. Write a recursive 'C' code to solve this problem and trace it for n= 3 disks.	10	L3	3	2
		Unit – IV				
7.	a)	Define Binary search tree. Write an algorithm to insert an item into a binary search tree.	10	L3	4	2
	b)	Briefly explain the various methods of implementing binary trees in C.	10	L2	4	1



8. a)	Write Prim's minimum spanning tree algorithm. Explain with example.	10	L3	4	2
b)	Develop an algorithm for Depth First Search in Graph. Trace it using Stack.	10	L3	4	2

Unit – V

9. a)	Write and explain Quick sort technique with example.	10	L3	5	2
b)	Compare and contrast Sequential search and Indexed Sequential search.	10	L2	5	1
10. a)	Explain the Merge sort technique. Sort the string MCADEPARTMENT in alphabetical order using this technique.	10	L3	5	2
b)	Compare the following searching techniques. i) Binary Search ii) Interpolation Search	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
